

Web Development — Final Project

PERSONAL TRIP PLANNER

By Offir Tura & Liad nave

APPLICATION INTERFACES

User Flow: Login → Plan → Map → Save/History


Welcome back
Sign in to plan your next adventure

[Sign in](#)

[No account? Create one](#)

[Plan](#) [History](#)

Plan your trip
Pick a country, choose hiking or biking, then generate a realistic route starting today.

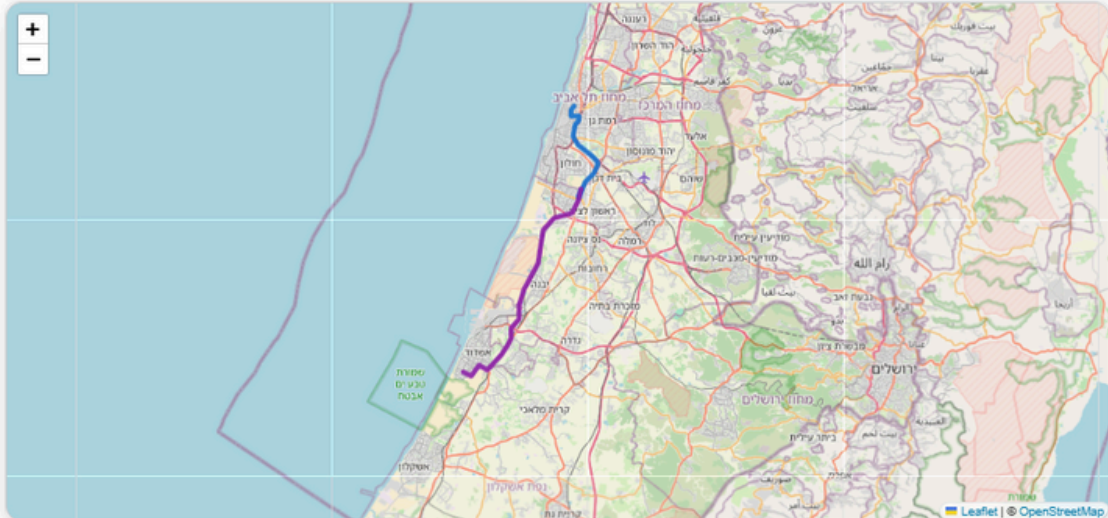
Country

☐ Hiking ☒ Biking

[Generate Real Route](#)

Total distance
49.11 km

Type
biking



[Plan](#) [History](#)

Your Trips
Find saved routes, filter by type, and open details.

Type

Search (name / description / weather)



Israel hiking trip
Loop Start near Jaffa Old City Loop Start near Mamilla Mall
Type: hiking · Distance: 25.6 km
[View](#) [Delete](#)



Netherlands biking trip
Day 1: Haarlem → Leiden Day 2: Leiden → Delft
Type: biking · Distance: 63.2 km
[View](#) [Delete](#)



UK hiking trip
Buckingham Palace and Tower of London
Type: hiking · Distance: 25.3 km
[View](#) [Delete](#)

ARCHITECTURE & DATA FLOW

Frontend

- React 18 for dynamic UI
- Leaflet.js + OpenStreetMap for interactive maps & markers
- Axios + AuthContext for API calls and session handling
- CSS Modules for modular styling

Database

- MongoDB (Mongoose) for flexible JSON-like storage
- Collections: users, trips (with waypoints)

Backend

- Node.js and Express.js for REST API and business logic
- Validation, retries, and error handling around LLM/routing calls

APIs & Auth

- Groq (LLM, llama-3.1) for POI generation
- OSRM for routing
- OpenWeatherMap for forecasts
- Unsplash for images
- JWT & bcrypt for secure authentication

```
1  PORT=5000
2
3  GROQ_API_KEY=*****
4  GROQ_MODEL=llama-3.1-8b-instant
5  GROQ_BASE_URL=https://api.groq.com/openai/v1
6
7  OSRM_BASE=https://router.project-osrm.org
8
9  MONGO_URI=mongodb+srv://*****:*****
10
11  JWT_SECRET=*****
12  WEATHER_API_KEY=*****
13
14  UNSPLASH_ACCESS_KEY=*****
```

.env file

LLM: POIS & GUARDRAILS

- Uses a free Groq LLM.
- LLM proposes named POIs/waypoints inside the selected country.
- Sometimes invalid coordinates occur (e.g., in the sea).
- We retry up to 5 times on infeasible routes: else show a clear error.

```
const system = {
  role: 'system',
  content:
    'You are a precise trip planner. Return STRICT JSON only (no extra text). ' +
    'Use well-known, routable places inside the specified country. ' +
    'For walking, choose diverse, named urban POIs (museums, art galleries, markets/bazaars, ' +
    'food halls, historic squares/plazas, old city gates/clock towers, libraries, universities/campuses, ' +
    'city halls/courthouses, theaters/opera houses/concert halls, cultural centers, religious sites ' +
    '(synagogues/churches/mosques/temples), monuments/statues/memorials, landmark bridges, observation ' +
    'decks/viewpoints, waterfront promenades/piers, central train/bus stations, iconic hotels, stadiums/arenas, ' +
    'science centers/aquariums/zoos, botanical gardens/greenhouses, covered arcades/passages, street-art alleys ' +
    'notable neighborhoods/streets), not only parks. ' +
    'All coordinates must be plausible decimal lat/lon inside the country.',
};
```

REAL PATHS: OSRM

- Snap and route through OSRM/ORS to ensure real, navigable paths.
- Hiking loop rule: start and end must be the same point.
- Check distance limits and inside-country bounds.
- On failure: retry; else show a clear error to the user.

Request

```
X Headers Payload Preview Response Initiator Timing
▼ Request Payload View source
{country: "Israel", tripType: "bike", limits: {min: 0, max: 60, maxDays: 2, loop: false}}
country: "Israel"
limits: {min: 0, max: 60, maxDays: 2, loop: false}
loop: false
max: 60
maxDays: 2
min: 0
tripType: "bike"
```

Response

```
{...}
days: [{day: 1, waypoints: [{name: "Tel Aviv", lat: 32.085329, lon: 34.785911},...], profile: "bike",...}]
  0: {day: 1, waypoints: [{name: "Tel Aviv", lat: 32.085329, lon: 34.785911},...], profile: "bike",...}
    day: 1
    distanceKm: 15.55
    durationMin: 17.5
    geometry: {...}
    profile: "bike"
    waypoints: [{name: "Tel Aviv", lat: 32.085329, lon: 34.785911},...]
  1: {day: 2, waypoints: [{name: "Rishon LeZion", lat: 31.987306, lon: 34.774386},...], profile: "bike",...}
    day: 2
    distanceKm: 33.89
    durationMin: 30
    geometry: {...}
    profile: "bike"
    waypoints: [{name: "Rishon LeZion", lat: 31.987306, lon: 34.774386},...]
```


PLAN YOUR TRIP WITH AI THAT KNOWS THE WAY

Plan your trip

Pick a country, choose hiking or biking, then generate a realistic route starting today.

Country

Israel

☒ Hiking ☐ Biking

Generate Real Route

Total distance
0.00 km

Type
hiking



Your Trips

Find saved routes, filter by type, and open details.

Type

All

Search (name / description / weather)

e.g. Italy, sunny, family



Israel hiking trip

Loop Start near Jaffa Old City Loop Start near Mamilla Mall

Type: hiking - Distance: 25.6 km

View

Delete



Netherlands biking trip

Day 1: Haarlem → Leiden Day 2: Leiden → Delft

Type: biking - Distance: 63.2 km

View

Delete



UK hiking trip

Buckingham Palace

Type: hiking - Distance: 10.0 km

View

Delete

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THANK YOU

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www.reallygreatsite.com